ACRME Calculation Logic Reference — All Scenarios (v2.2)

Revision v2.2 — 27 August 2026. This document supersedes the previous calculation logic reference. Key changes in v2.2: - **DR sizing formula replaced**: fixed `dr_ratio_*` percentage approach (v1) is superseded by the **max-not-sum distributed-portion formula** (Requirements v2.1 Appendix A.6 / Appendix D). - **Reservation target floor formula added**: `Allocated VM Count + Configured Buffer` (CAP-003), distinct from the forecast-based growth formula. - **Four new scenarios added**: DR destination requirement (max-not-sum), standby activation, deployment readiness gate, and customer seed record. - **EU cross-geo DR region corrected**: Belgium Central → **Switzerland North** throughout. - **Scenario 15** (fixed DR ratio parameters) retained for reference but marked **superseded**.

This document consolidates **every calculation logic** used by the Azure Capacity Reservation Management Engine (ACRME), organized by the scenario in which it fires. Each formula carries an evidence tag:

- `[Documented]` — backed by Microsoft Learn or Azure platform documentation.
- `[Decided]` — an approved design decision recorded in the ADR Decision Log.
- `[Derived]` — a logical consequence of the design; requires POC validation where noted.
- `[Assumed]` — a policy default or working hypothesis, tunable and not yet empirically validated.

Sources: Requirements Baseline v2.1/v2.2 (Appendix A–D); ADR-001 through ADR-004 v1.2; `acrme_production_readiness_review_and_architecture.md` (PRR §26–§32).

Convention. `vCPU` = `vCPU_per_instance` for the SKU family. `CVAL` (Customer Validation) and `NonProd` are interchangeable; scoring identifiers keep the `NonProd` spelling for engineering consistency. All raw ratios are clamped with `Clamp(x) = max(0, min(1, x))` before weighting.

Index of Scenarios

#	Scenario	Core calculation	Status
1	Prod region derivation — customer picks a **geography** (**exception path**; explicit approval + customer acknowledgement required)	`argmax(PS_Prod)` over Standard regions	Current
2	Prod region validation — customer supplies the **exact region** (**default input**; ACRME validates, does not derive)	HC-1..HC-10 gate + `PS_Prod` post-validation	Current
3	Restricted region request	Exception workflow (no scoring)	Current
4	Middle East three-region deployment	`argmax(PS_Prod)` in-geo + **Switzerland North** cross-geo DR	**Updated**
5	CVAL / NonProd region selection	`argmax(PS_NonProd)`	Current
6	DR region selection	`argmax(PS_DR)`	Current
7	Hard-constraint eligibility gate	HC-3, HC-6, HC-7 arithmetic	Current

#	Scenario	Core calculation	Status
8	Quota group sizing	Prod & NonProd+DR group budgets	Current
9	DR floor accounting	`DR_Floor_vCPU`, effective ceiling, headroom	**Updated**
10	Environment quota scoring	`Quota_Score_{Prod,NonProd,DR}`	Current
11	Capacity forecast / CR sizing	`Forecast_Quantity`	Current
12	Auto-increase trigger	Utilisation thresholds + debounce	Current
13	Emergency transfer & tier escalation	Tier 1/2/3, quota-neutral transfer	Current
14	Scaling & API-budget model	CRG cardinality, request-per-cycle	Current
15	DR ratio parameters (fixed %)	`dr_ratio_min/max/target`	**Superseded by Scenario 17**
16	Reservation target & reconciliation floor	`Allocated VM Count + Buffer` (CAP-003)	**New**
17	DR destination requirement — max-not-sum	`MAX(source portions per destination)` (A.6)	**New**
18	Standby activation (DR-019)	`associated → allocated` staged acquisition	**New**
19	Deployment readiness gate	`READY / QUOTA_DEFICIT / STALE_STATE / …` (RDY-002)	**New**
20	Customer placement seed record	First-placement persistence + reuse policy	**New**

Shared Building Blocks (used by several scenarios)

Default scoring weights `[Assumed]`

```
alpha ( ) = 0.30   # capacity / headroom signal
beta  ( ) = 0.20   # quota headroom signal
gamma ( ) = 0.25   # capacity-weighted distribution
delta ( ) = 0.15   # DR-coverage / overflow-integrity signal
epsilon( )= 0.10   # zone diversity
Clamp(x)  = max(0, min(1, x))       # every component clamped to [0,1] before weighting
```

All five weights are tunable policy constants stored in `PlacementPolicy` (config-as-code, versioned). They sum to exactly 1.0 and apply across all three environment scoring formulas. `[Assumed]`

Named regional-snapshot quantities `[Decided]`

```
nonprod_crg.effective_free = nonprod_crg.free_slots - dr_overflow_reserve
dr_crg.coverage_ratio      = dr_crg.quantity / potential_dr_demand
potential_dr_demand(region)= Σ prod_allocated  for all customers whose dr_region = region
```

Key formula disambiguation

Formula	Purpose	Source
`Target Reserved Capacity = Allocated + Buffer`	Continuous reconciliation floor (Scenario 16)	CAP-003
`Forecast_Quantity = ceil(Peak × (1+Growth) + DR_Buffer)`	Proactive growth ahead of demand (Scenario 11)	ADR-004
`Destination DR Requirement(d) = MAX(source portions)`	DR standby sizing (Scenario 17)	A.6 / DR-017
`dr_ratio_*` constants	Superseded (Scenario 15) — configuration reference only	PRR §32

Scenario 1 — Prod Region Derivation (Customer Chooses a Geography)

Trigger: customer supplies a geography (e.g. “Europe”), not a specific region.
Status: Exception path — not the default (PLC-002). Requires explicit exception approval and customer acknowledgement that the **derived production region becomes fixed until an approved migration changes the seed**. Cannot be invoked without a recorded exception reference. `[Decided]`

Logic:

1. Build candidate set = **Standard Capacity Regions** in the chosen geography. Restricted regions are excluded before scoring. `[Decided]`
2. Score every candidate with `PS_Prod` from the versioned regional snapshot.
3. Select `derived Prod anchor = argmax(PS_Prod)`.
4. **Tie-break:** deterministic — first region in the Standard Capacity Region list order for that geography; also the cold-start default when no snapshot exists. `[Derived]`
5. Run sequential CVAL → DR selection (Scenarios 5 & 6) on the same Standard region pool.

`PS_Prod(r) [Decided]`

```
PS_Prod(r) =
    0.30 × Clamp(nonprod_crg.effective_free / prod_crg.quantity)     ← : NonProd headroom signal
  + 0.20 × Clamp(prod_crg.quota_headroom  / prod_crg.quota_limit)    ← : Prod quota headroom
  + 0.25 × Clamp(1 - prod_customer_count  / total_customers)         ← : distribution fairness
  + 0.15 × Clamp(dr_crg.coverage_ratio)                              ← : DR readiness signal
  + 0.10 × Clamp(az_count / 3)                                       ← : zone diversity
```

`PS_Prod` is **dual-purpose**: derives the Prod region in Scenario 1 and is reused for post-selection validation in Scenario 2. Every candidate score and the policy version are written to the `OperationRecord` for deterministic replay. `[Derived]`

Scenario 2 — Prod Region Validation (Customer Supplies a Specific Region)

Trigger: customer names an exact Azure region — the **default** input mode (PLC-001). ACRME **validates** the supplied region against HC-1..HC-10 and placement-policy rules; it does **not** derive a region from geography. `[Decided]`

Logic:

- **Standard Capacity Region:** validate against HC-1..HC-10 (Scenario 7). If eligible, it becomes the Prod anchor directly. `PS_Prod` computed for post-selection validation/audit only, not for selection.
- **Restricted Capacity Region:** do not score — route to the Exception Deployment Workflow (Scenario 3).

There is no `argmax` here. The customer's choice is authoritative once it passes the hard constraints. The first approved placement writes the **Customer Seed Record** (Scenario 20). `[Decided]`

Scenario 3 — Restricted Region Request (Exception Path)

Trigger: the requested or only feasible region is a **Restricted Capacity Region**.

Logic: no scoring formula is evaluated. The request diverts to the manual Exception Deployment Workflow (governance approval, EC-1..EC-4, VR-1..VR-11, HC-9/HC-10 controls). Restricted regions never enter the scoring pipeline. `[Decided]`

Exception conditions:

Code	Condition
EC-1	Customer requires the specific restricted region by contract
EC-2	No Standard region in the geography can supply sufficient capacity
EC-3	Data-residency requirement binds to the restricted region
EC-4	Scenario 2 input — restricted regions can never be engine-derived

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Scenario 4 — Middle East Three-Region Deployment (Updated — Switzerland North)

Trigger: a Middle East deployment requiring three environments.

Logic:

```
1. Candidate in-geo Standard regions = { Saudi Arabia Central, UAE North }   (both Standard)
2. Score both with PS_Prod.
3. Prod  = argmax(PS_Prod) over the two.
4. CVAL  = the remaining in-geo region (deterministic - only one candidate left).
5. DR    = cross-geo extension region: Switzerland North (Europe)
             because no third in-geo Standard region exists to satisfy region separation.
```

> **v2.2 correction (REG-002):** The cross-geo DR extension region was incorrectly cited as "Belgium Central" in earlier material. The authoritative placement configuration specifies **Switzerland North** as the cross-geo extension for Middle East. All references to Belgium Central are superseded by Switzerland North.

Middle East is subject to the `DR_NOT_OFFERED` policy flag (DR-014) where legal/data-sovereignty prevents an acceptable DR design. When `DR_NOT_OFFERED = true` for the geography, Step 5 above is bypassed and the seed record records `dr_region = NOT_OFFERED`. `[Decided]`

Cross-geo extension constraints from ADR-001 apply to the DR region (Switzerland North). `[Decided]`

Scenario 5 — CVAL / NonProd Region Selection

Trigger: sequential step after the Prod anchor is fixed. Selects `argmax(PS_NonProd)` over eligible Standard regions.

`PS_NonProd(r)` — design-of-record `[Decided]`

```
PS_NonProd(r) =
    0.30 × Clamp(nonprod_crg.effective_free / nonprod_crg.quantity)      ← : effective NonProd headroom
  + 0.20 × Clamp(nonprod_crg.quota_headroom / nonprod_crg.quota_limit)   ← : NonProd quota headroom
  + 0.25 × Clamp(1 - nonprod_customer_count / total_customers)           ← : distribution fairness
  + 0.15 × Clamp(nonprod_crg.effective_free / nonprod_crg.quantity)      ← : overflow capacity health
  + 0.10 × Clamp(az_count / 3)                                           ← : zone diversity
```

Corrected pilot variant `[Undocumented - architectural judgement]`

The design-of-record above duplicates the same signal under and (combined weight 0.45). The PRR's corrected pilot formula removes the duplication:

```
PS_NonProd = 0.35 × Capacity + 0.25 × Quota + 0.25 × Distribution
           + 0.05 × DR_Overflow_Integrity + 0.10 × Zones

Distribution = 1 - Region_Assigned_Demand / Total_Assigned_Demand
```

Both variants are recorded. The design-of-record formula is the approved baseline; the pilot variant is the reviewer-recommended refinement. Neither is empirically validated yet.

CVAL/DR co-location rule (PLC-010): a customer's CVAL and DR *may* co-locate in the same destination region. When co-located, CVAL capacity earmarked for DR activation must **not** be double-counted as both live CVAL headroom and available DR headroom. The `CVALEarmarkRecord` tracks this. `[Decided]`

Scenario 6 — DR Region Selection

Trigger: final sequential step. Selects `argmax(PS_DR)` over eligible Standard regions (or the cross-geo region for Middle East).

`PS_DR(r) [Decided]`

```
PS_DR(r) =
    0.30 × Clamp(dr_crg.free_slots / dr_crg.quantity)                  ← : DR CRG headroom
  + 0.20 × Clamp(dr_crg.quota_headroom / dr_crg.quota_limit)          ← : DR quota headroom
  + 0.25 × Clamp(1 - dr_customer_count / total_customers)             ← : distribution fairness
  + 0.15 × min(1.0, dr_crg.coverage_ratio / dr_ratio_target)          ← : coverage-ratio health
  + 0.10 × Clamp(az_count / 3)                                        ← : zone diversity
```

Note on : `dr_ratio_target` in the PS_DR formula is now a **configurable bootstrap reference** rather than a fixed 30–40% constant. See Scenario 17 for the authoritative DR sizing formula. `[Decided]`

Scenario 7 — Hard-Constraint Eligibility Gate (Arithmetic Constraints)

Before any region is scored (Scenario 1) or accepted (Scenario 2), it must pass the hard constraints.

HC-3 QUOTA_FLOOR (per environment) `[Decided]`

```
Prod:    Prod_Group_headroom(R)    requested_vm_count × vCPU
NonProd: nonprod_headroom(R)       requested_vm_count × vCPU    # uses effective_nonprod_ceiling
DR:      dr_headroom(R)            target_dr_qty      × vCPU
```

HC-6 DR_COVERAGE_FLOOR `[Decided]`

```
Region ineligible for DR if:
    (dr_crg.free_slots + nonprod_crg.effective_free) < customer_requested_dr_slots
```

HC-7 DR_FLOOR_INTEGRITY `[Decided]`

```
REJECT NonProd placement in region R if:
    nonprod_quota_used(R) + (requested_vm_count × vCPU) > effective_nonprod_ceiling(R)
```

HC-3 checks raw group headroom; HC-7 additionally enforces floor compliance. Both must pass. On breach: emit `DRFloorViolationDetected` (Critical severity).

Minimum-headroom policy defaults `[Decided]`

```
min_prod_quota_headroom_vcpu    = 20
min_nonprod_quota_headroom_vcpu = 20
min_dr_headroom_vcpu            = 16
```

Scenario 8 — Quota Group Sizing (per Region)

Every managed region has exactly two Azure Quota Groups. `[Decided]`

```
Prod_Group_Limit(region) =
    Prod_CRG_quantity × vCPU × (1 + prod_growth_buffer)
    prod_growth_buffer = 0.20

NonProd_DR_Group_Limit(region) =
      NonProd_CRG_quantity × vCPU × (1 + nonprod_growth_buffer)
    + DR_CRG_quantity      × vCPU
    + emergency_transfer_headroom_vcpu

nonprod_growth_buffer          = 0.20
emergency_transfer_headroom_vcpu = max_emergency_transfer_qty × vCPU
max_emergency_transfer_qty       = potential_dr_demand × emergency_transfer_pct
emergency_transfer_pct           = 0.30
```

The `emergency_transfer_headroom_vcpu` term is what makes Tier 2 DR expansion quota-neutral within the shared group. `[Decided]`

Quota-as-governor principle (QUA-005): quota allocation caps deployable capacity per product, environment, subscription, region, and VM family. Teams must justify quota increases — unallocated pooled quota is not automatically distributed. `[Decided]`

Scenario 9 — DR Floor Accounting (Engine-Enforced Sub-Limit) — Updated

Azure Quota Groups have no native intra-group sub-reservation; the engine enforces the DR floor by arithmetic. `[Decided]`

Accounting formulas [Decided]

```
DR_Floor_vCPU(region)      = Destination_DR_Requirement(region) × vCPU
                             [v2.2: uses max-not-sum formula from Scenario 17]
                             [v1: was potential_dr_demand × vCPU × dr_ratio_max - superseded]

Effective_NonProd_Ceiling = NonProd_DR_Group_Limit - DR_Floor_vCPU
NonProd_Headroom          = Effective_NonProd_Ceiling - NonProd_Used_vCPU
Group_Headroom            = Group_Limit - Group_Used
dr_crg.coverage_ratio     = dr_crg.quantity / Destination_DR_Requirement(region)
```

> **v2.2 change:** DR_Floor_vCPU is no longer derived from potential_dr_demand × dr_ratio_max. It is now derived from Destination_DR_Requirement(region) — the max-not-sum value from Scenario 17. This aligns the floor with the lean bootstrap model (DR-007) and the distributed DR topology (DR-016). The fixed dr_ratio_max = 0.40 constant is retained in Scenario 15 for legacy reference only.

Dual-validation control [Decided]

A separate detector recomputes the DR floor from authoritative assignment data; any disagreement between command-time and detector values disables automatic NonProd expansion until reconciled.

Worked topology example [Decided]

```
Prod group          = 128 vCPU  (32 × D4s_v3, 4 vCPU each)
NonProdDR group     =  80 vCPU
DR floor            =  32 vCPU   (= Destination_DR_Requirement for this region)
Effective NonProd ceiling = 48 vCPU  (80 - 32)
```

Scenario 10 — Environment Quota Scoring (Component Detail)

The generic Quota_Score is dispatched by environment type. [Decided]

```
Quota_Score_Prod(r)    = prod_group_headroom_vcpu   / prod_group_limit_vcpu
Quota_Score_NonProd(r) = nonprod_headroom_vcpu       / effective_nonprod_ceiling_vcpu
Quota_Score_DR(r)      = dr_headroom_vcpu            / dr_floor_vcpu
```

Semantics differ by type: a DR score of 1.0 means maximum expansion room (DR CRG at zero), not maximum quota consumed. Zero-denominator guards apply. [Decided]

Scenario 11 — Capacity Forecast / CR Sizing

Advisory forecast used to size CR quantity over a horizon — the **proactive growth path**, distinct from the continuous reconciliation floor (Scenario 16). [Decided]

```
Forecast_Quantity = ceil( Forecast_Peak × (1 + Growth_Buffer) + DR_Buffer )

Forecast_Peak    = predicted peak allocated-VM demand in the horizon          [Derived]
Growth_Buffer    = policy % for forecast uncertainty                          [Assumed]
DR_Buffer        = extra units required by approved recovery policy           [Assumed]
Forecast_Horizon  { 30, 60, 90 } days                                        [Assumed]
```

Lead-time alerting: when forecast demand approaches 80% of quota limit, emit ForecastApproachingQuotaLimit with 14-day lead time. [Documented]

Forecast recommendations remain advisory until model accuracy and false-positive rates are measured. The horizon and buffer values are policy percentages stored in `PlacementPolicy`, not fixed constants.

Scenario 12 — Auto-Increase Trigger (Steady-State Capacity Lifecycle)

When utilisation crosses a threshold, a `CapacityIncreaseRequest` is raised. `[Decided]`

```
Trigger auto-increase when utilisation  threshold:
    dr_autoincrease_threshold          = 0.35
    prod_autoincrease_threshold        = 0.20
    nonprod_autoincrease_threshold     = 0.20

Debounce cooldown = 30 min per (region + CRG type) after any trigger
```

Phase 1 requires operator approval. Auto-decrease excluded from Phase 1. The steady-state lifecycle runs **only** in `STEADY_STATE` engine mode (see ADR-003). `[Decided]`

Normative 10-step steady-state capacity lifecycle

1. Detect threshold crossing.
2. Re-read current CR, quota, sharing, and assignment state.
3. Create `CapacityIncreaseRequest`.
4. Calculate target quantity (`Forecast_Quantity` or `Allocated + Buffer` floor, whichever is higher).
5. Require operator approval (Phase 1).
6. Submit the quota action only if validated as required.
7. Wait for confirmed quota state — no assumed propagation SLA.
8. Update CR quantity.
9. Confirm the actual quantity.
10. Refresh the snapshot and close the request.

Scenario 13 — Emergency Transfer & Tier Escalation (DR Event Active)

During an active DR event (`DR_EVENT_ACTIVE` engine mode), emergency capacity is obtained by tier. `[Decided]`

```
Tier 1  DirectExpansion       - additive DR expansion using existing headroom. Automated.
Tier 2  QuotaNeutralTransfer  - reduce NonProd CR, expand DR within the SAME quota group.
                                Policy-gated. Quota-neutral (see below).
Tier 3  DestructiveTransfer   - changes VM associations. Dual-approval + elevated RBAC.
                                BLOCKED in Phase 1.
```

Quota-neutral math for Tier 2 `[Derived - POC-31/32]`

```
Because NonProd CR and DR CR draw from the SAME NonProd+DR quota group:
    NonProd CR reduction → releases q × vCPU to the GROUP pool
    DR CR expansion      → consumes q × vCPU from the SAME GROUP pool
    Net group headroom change  0 → no Azure quota-increase request needed
    Tier RTO gated only by ARM operation time (minutes), not quota approval (hours)

max_emergency_transfer_qty = potential_dr_demand × emergency_transfer_pct  (pct = 0.30)
```

Escalation order: `Tier 1 headroom available? → Tier 2 approved? → Tier 3 allowed? → manual`.

Scenario 14 — Scaling & API-Budget Model

Used to size the estate and stay within Azure Resource Manager throttling budgets. `[Derived]`

```
CRG_lower_bound = 3 × R                     # 3 CRGs (Prod/NonProd/DR) per region
CRG_total       = R × Eclass × Z × SKUset × IsolationFactor

Requests_per_cycle = CRG_reads + CR_reads + sharing_reads + quota_reads + association_reads
Average_requests_per_second = Requests_per_cycle / 300          # 5-minute (300 s) cycle
```

Documented ARM throttle baselines `[Documented]`

```
Compute RP read  limit = 250   requests / 5 min / subscription
Compute RP write limit = 1,200 requests / hour  / subscription
```

The adaptive throttle manager initialises with these baselines and adapts to observed `429/Retry-After`.

Reconciliation cadence `[Assumed]`

```
Reconciliation loop target       = 6 min (configurable; production interval TBD)
Critical targeted reconcile P95  < 2 min
Stable-resource state age  P95   < 15 min
Drift detection                    2 reconciliation cycles
```

Scenario 15 — DR Ratio Parameters (Fixed %) — SUPERSEDED

> **Status: Superseded by Scenario 17 (max-not-sum).** These constants are retained for configuration reference and legacy comparison only. The fixed-percentage DR sizing model (`30-40% of Prod`) was rejected in Requirements v2.0/v2.1 because it produces idle reserves estimated at **$1.5M–$5M/year** at platform scale — directly contradicting the cost-reduction mandate. See Appendix D of the Requirements Baseline for the full derivation.

```
dr_ratio_min    = 0.30      [superseded - was HC-6 lower bound]
dr_ratio_max    = 0.40      [superseded - was DR_Floor_vCPU multiplier]
dr_ratio_target = [0.30, 0.40]   [superseded - now: configurable bootstrap per product/workload]
emergency_transfer_pct = 0.30    [still current - Scenario 13]
prod_growth_buffer     = 0.20    [still current - Scenario 8]
nonprod_growth_buffer  = 0.20    [still current - Scenario 8]
```

The `SUM` override (C-11 in the Configurable Items Register) remains available for specific customers or geographies that contractually require protection against concurrent regional failures. All other scopes use the max-not-sum formula (Scenario 17).

Scenario 16 — Reservation Target & Reconciliation Floor (CAP-003) — New

Trigger: every reconciliation cycle compares current reserved quantity against this floor. This is the **continuous floor formula** — distinct from the forecast-based growth formula (Scenario 11).

A.1 Reservation target (CAP-003) `[Decided]`

```
Target Reserved Capacity = Allocated VM Count + Configured Buffer
```

- `Allocated VM Count` = VMs in `running / allocated` state consuming compute capacity.
- `Configured Buffer` = policy-defined headroom above allocated demand. Tunable per product/region/env/SKU. Not hard-coded.

- **Associated-but-deallocated VMs** do not force reservation retention (CAP-004). They are reported separately so teams understand restart risk, but do not add to the target.

A.2 Reservation headroom

```
Reservation Headroom = Reserved Quantity - Allocated VM Count
```

A.3 Reservation deficit

```
Reservation Deficit = max(0, Target Reserved Capacity - Reserved Quantity)
```

Reconciliation decision logic `[Decided]`

```
IF Reserved Quantity < Target Reserved Capacity:
    RAISE reservation (scale-up) toward target, subject to Azure availability.
    IF Azure cannot supply: hold current state, RAISE alert, expose buffer deficit.

IF Reserved Quantity > Target Reserved Capacity (consistently, after minimum-hold interval):
    LOWER reservation toward target - ONLY when all guards pass:
        - current_reserved > allocated_count         (never below allocated)
        - not within DR protection window
        - no active maintenance exclusion
        - cost policy permits reduction
```

Zero-capacity and no-auto-delete policy (CAP-009/010) `[Decided]`

```
Reduce an unused managed reservation to ZERO; do NOT delete the reservation object.
Normal reconciliation NEVER deletes CRGs or reservation definitions.
Deletion uses a separate approved decommissioning workflow only.
```

Scenario 17 — DR Destination Requirement — Max-Not-Sum (A.6/DR-017) — New

This scenario replaces the fixed `dr_ratio_*` sizing in Scenario 15.

The single-failure assumption (DR-001) `[Decided]`

The programme plans for failure of **one** production region within a geography at a time. Simultaneous multi-region failure is outside the default guaranteed model.

A.6 DR destination requirement — corrected formula `[Decided]`

```
Destination_DR_Requirement(d)
  = MAX over each non-concurrent source region s protected by d (
        Workload_Portion(s → d)
    )
```

Where: - `d` = destination region hosting DR standby capacity - `s` = source region whose production customers have DR in `d` - `Workload_Portion(s → d)` = the quantity of VMs/vCPUs from source `s` that are assigned to land in destination `d` on failover (from the source→destination DR index, DR-018)

Rationale: because only one region fails at a time, destination `d` never simultaneously hosts failover from more than one source. It therefore needs standby capacity for its **largest** protected source only — never the sum of all sources. The standby slots are **shared / overcommitted** across mutually exclusive failure events. `[Decided]`

A.7 DR capacity gap `[Decided]`

```
DR_Capacity_Gap(d) = max(0,
    Destination_DR_Requirement(d) - Usable_Destination_Capacity(d)
)
```

`Usable_Destination_Capacity(d)` may include approved bootstrap headroom, available CRG reservations, releasable CVAL capacity (after earmark check, Scenario 9/PLC-010), and capacity acquired through approved sharing or expansion.

A.8 Shared-DR overcommit ratio (informational) `[Derived]`

```
Overcommit_Ratio(d) = SUM(Workload_Portion(s → d) for all s) / MAX(Workload_Portion(s → d))
```

A ratio > 1 quantifies: (a) the capacity and cost saved by max-not-sum sizing, and (b) the exposure if the single-failure assumption is ever violated. Required input for POC-011 risk sign-off.

Observability requirement (OBS-002) `[Decided]`

Alert when a destination's usable capacity falls below `Destination_DR_Requirement(d)` — the max protected source — not below the sum. Dashboard must show per-destination max-source coverage (OBS-004).

Conservative SUM override (C-11) `[Decided]`

```
Destination_DR_Requirement_Conservative(d) = SUM(Workload_Portion(s → d) for all s)
```

Used only where a customer or geography contract explicitly requires protection against concurrent regional failures. Configured per-scope in `PlacementPolicy`; no code change required.

Worked example (from Requirements Appendix D) `[Decided]`

Region 2 holds DR standby for customers from Region 1 and Region 3:

Source protected by R2	Failover portion in R2
Region 1 (Cust1 + Cust5)	120 cores
Region 3 (Cust6)	80 cores

```
SUM sizing (old):  Requirement(R2) = 120 + 80 = 200 cores   ← over-provisions; superseded
MAX sizing (new):  Requirement(R2) = max(120, 80) = 120 cores

Saving at R2: 200 → 120 = 80 cores (40%) removed with no loss of protection.

Overcommit_Ratio(R2) = 200 / 120   1.67
→ R2's shared standby is 1.67× oversubscribed.
→ Platform-wide extrapolation: ~40-50% standby reduction   $0.6M-$2.5M/year saved.
```

Source→Destination DR Index entity (DR-018) `[Decided]`

The max-not-sum formula depends on an authoritative bidirectional mapping — a required state-store entity:

```
SourceDestinationDRIndex {
    source_region          : string
    destination_region     : string
    customer_realm_id      : string
    standby_instance_set   : list<InstanceRef>
    sku_family             : string
    quantity_vcpu          : int
```

```
    activation_state       : STANDBY | ACTIVATING | ACTIVE | FAILBACK_PENDING
    last_updated           : datetime
    policy_version         : string
}
```

This index is the reverse view of the customer seed record (Scenario 20). On a regional failure, the engine queries this index to determine exactly which standby sets to activate and where. `[Decided]`

Scenario 18 — Standby Activation (DR-019) — New

Trigger: authorised DR declaration; engine mode transitions to `DR_EVENT_ACTIVE`.

Per-customer activation workflow `[Decided]`

Each customer's DR instances transition `associated → allocated` (standby → active) in approved business-priority waves:

```
Priority Wave 0 (P0):   Platform control planes and recovery orchestration infrastructure.
Priority Wave 1 (P-1):  Highest-priority customer workloads.
Priority Wave N:        Remaining customers in business-priority order.
```

DR-006 staged capacity acquisition per wave:

```
Stage 1: Use approved bootstrap / pre-staged headroom (zero Azure wait time).
Stage 2: Allocate available reservation + quota already in the destination.
Stage 3: Shut down / disassociate eligible CVAL workloads (after DR-010 authorised trigger).
Stage 4: Share or reassign reservations within supported region/zone boundaries.
Stage 5: Allocate pooled quota to DR subscriptions.
Stage 6: Request additional Azure quota/capacity where required.
Stage 7: Report unrecoverable capacity gaps.
```

Stages 1–2 start immediately from bootstrap headroom — P0/P-1 customers begin recovery without waiting on Azure deallocation/disassociation APIs (which are throttled during regional events). `[Decided]`

Activation state tracking `[Decided]`

```
ActivationRecord {
    customer_realm_id   : string
    source_region       : string
    destination_region  : string
    priority_wave       : int
    activation_state    : PENDING | ACTIVATING | ACTIVE | FAILED | FAILBACK_PENDING
    stage_reached       : int           # 1-7 per DR-006
    capacity_gap_vcpu   : int           # 0 if fully provisioned
    activated_at        : datetime
    approved_by         : string
    incident_id         : string
}
```

Activation state is tracked per customer, auditable, and reversible on failback (DR-013). `[Decided]`

Failback reversal `[Decided]`

On failback: `ACTIVE → FAILBACK_PENDING → STANDBY` in reverse priority order. The seed record and DR index are preserved; no re-derivation of placement. History, audit trail, and DR capacity state are retained through the full failback cycle.

Scenario 19 — Deployment Readiness Gate (RDY-002) — New

Trigger: every AEP-triggered deployment before the first environment is provisioned.

Machine-readable readiness states `[Decided]`

```
READY                  All gates pass; deployment may proceed.
READY_WITH_RISK        All gates pass but one or more advisory risks detected (e.g. buffer below target)
QUOTA_DEFICIT          Required deployment quota > available quota in the deploying subscription.
RESERVATION_DEFICIT    Reservation exists but quantity insufficient; approved over-allocation not in eff
CAPACITY_UNAVAILABLE   Azure cannot supply the requested SKU/zone at this time.
STALE_STATE            Capacity/quota snapshot exceeds max-age threshold; cannot drive a safe placement.
POLICY_BLOCKED         A governance policy (hard constraint, exception, approval) blocks the deployment.
VALIDATION_REQUIRED    One or more validation steps (e.g., POC-gated behaviours) not yet confirmed.
```

Readiness gate logic (RDY-001) `[Decided]`

All of the following must pass for `READY`:

```
1. target_region     approved_region_catalogue         (REG-001)
2. az_count          1 in target_region                (CAP-011)
3. sku               supported_skus(target_region)
4. reservation_policy known for (subscription, region, zone, sku)
5. reservation exists (if required by policy)
6. reserved_quantity  requested_count OR approved_over_allocation active
7. consumer_subscription_quota  requested_units        (QUA-013 - POC-gated)
8. snapshot_age      max_snapshot_age_seconds           (RDY-004)
9. no blocking policy exception or hard constraint violation
```

Staleness check (RDY-004) `[Decided]`

```
IF now() - snapshot.collected_at > max_snapshot_age:
    RETURN STALE_STATE
    ACTION: refresh synchronously OR block with STALE_STATE - never drive placement from stale data
```

Quota deficit formula (A.5) `[Decided]`

```
Quota_Deficit = max(0, Requested_Deployment_Units - Available_Quota)
Available_Quota = Assigned_Regional_VM_Family_Quota - Current_Regional_VM_Family_Usage   (A.4)
```

Scenario 20 — Customer Placement Seed Record (PLC-003/004/005) — New

Trigger: first approved production-region placement for a customer in a geography.

Seed record creation `[Decided]`

```
CustomerSeedRecord {
    customer_realm_id       : string        # authoritative customer/realm identifier
    geography               : string        # e.g. "NorthAmerica", "Europe"
    production_region       : string        # exact Azure region name
    cval_region             : string        # exact Azure region name
    dr_region               : string | "NOT_OFFERED"
    products_covered        : list<string>  # all products seeded by this record
```

```
    decision_timestamp      : datetime
    policy_version          : string        # PlacementPolicy version used
    capacity_snapshot_ref   : string        # snapshot ID that drove the decision
    exception_ref           : string | null # approval ID if Restricted region or geo-exception
    approval_metadata       : ApprovalRecord
    input_mode              : "SPECIFIC_REGION" | "GEOGRAPHY_EXCEPTION"
}
```

Seed reuse policy (PLC-004) `[Decided]`

```
IF CustomerSeedRecord exists for (customer_realm_id, geography):
    READ seed record - do NOT re-invoke the placement engine.
    Subsequent products and environments use the seeded production/CVAL/DR regions.
    Engine is NOT re-run per product once seeded.
```

Controlled seed change (PLC-005) `[Decided]`

```
IF seed change requested:
    REQUIRE approved migration/exception workflow.
    REQUIRE impact analysis across all products covered by the seed.
    PROHIBIT automatic regeneration on: upgrades, rebuilds, routine deployments.
```

A. Core Formula Reference (Appendix A — Requirements v2.1)

Label	Formula	Source
A.1	`Target Reserved Capacity = Allocated VM Count + Configured Buffer`	CAP-003
A.2	`Reservation Headroom = Reserved Quantity - Allocated VM Count`	CAP-003
A.3	`Reservation Deficit = max(0, Target Reserved Capacity - Reserved Quantity)`	CAP-003
A.4	`Available Quota = Assigned Regional VM-Family Quota - Current Usage`	QUA-002
A.5	`Quota Deficit = max(0, Requested Deployment Units - Available Quota)`	QUA-007
A.6	`Destination_DR_Requirement(d) = MAX over s (Workload_Portion(s → d))`	DR-017
A.7	`DR_Capacity_Gap(d) = max(0, DR_Requirement(d) - Usable_Capacity(d))`	DR-017
A.8	`Overcommit_Ratio(d) = SUM(portions) / MAX(portions)`	DR-017

B. Consolidated Policy-Constant Table (v2.2)

Constant	Value	Used in Scenario(s)	Status
`alpha / beta / gamma / delta / epsilon`	0.30 / 0.20 / 0.25 / 0.15 / 0.10	1, 5, 6	Current
`prod_growth_buffer`	0.20	8	Current
`nonprod_growth_buffer`	0.20	8	Current
`emergency_transfer_pct`	0.30	8, 13	Current
`min_prod_quota_headroom_vcpu`	20	7	Current
`min_nonprod_quota_headroom_vcpu`	20	7	Current
`min_dr_headroom_vcpu`	16	7	Current
`dr_autoincrease_threshold`	0.35	12	Current
`prod/nonprod_autoincrease_threshold`	0.20	12	Current
Debounce cooldown	30 min	12	Current
Reconciliation loop target	6 min (configurable)	14	Current
ARM read / write baseline	250 per 5 min / 1,200 per hour	14	Current
`Forecast_Horizon`	30 / 60 / 90 days	11	Current
`dr_ratio_min`	0.30	15	**Superseded**
`dr_ratio_max`	0.40	15	**Superseded**
`dr_ratio_target`	[0.30, 0.40]	15	**Superseded**
DR bootstrap target	Configurable per product/workload — no fixed %	17	**Replaces ratio**
Max-not-sum default	`MAX(source portions)`	17	Current
SUM override (C-11)	`SUM(source portions)` — per-scope opt-in	17	Current
EU cross-geo DR extension region	**Switzerland North**	4	**Updated (was Belgium Central)**

C. Evidence & Maturity Summary

Formula	Evidence level	Validation required
`PS_Prod`, `PS_NonProd`, `PS_DR` (scoring)	`[Decided]` — design-of-record	Shadow/recommendation mode until empirically validated
HC-3, HC-6, HC-7 (hard constraints)	`[Decided]`	Ready for Phase 1
DR floor accounting (Scenario 9)	`[Decided]`	Updated to use max-not-sum input
Tier 2 quota-neutral math (Scenario 13)	`[Derived]`	POC-31/POC-32 required
Max-not-sum DR sizing (Scenario 17)	`[Decided]`	POC-011 required before production dependency
Consumer-subscription quota (QUA-013)	`[Assumed]`	POC-001 — **top technical unknown**
Reservation target floor (Scenario 16)	`[Decided]`	Ready for Phase 1
Standby activation staging (Scenario 18)	`[Decided]`	Dependent on POC-005 (VM state semantics)

Formula	Evidence level	Validation required
Deployment readiness gate (Scenario 19)	`[Decided]`	Ready for Phase 1
Customer seed record (Scenario 20)	`[Decided]`	Ready for Phase 3

All constants are policy defaults stored in `PlacementPolicy` (config-as-code, versioned). Tuning any constant requires updating the config; no code change is needed. The `dr_ratio_*` constants are retained in the codebase as fallback references for the SUM override (C-11) only.

Document version 2.2 — 27 August 2026. Next review: upon POC-001 / POC-011 results.